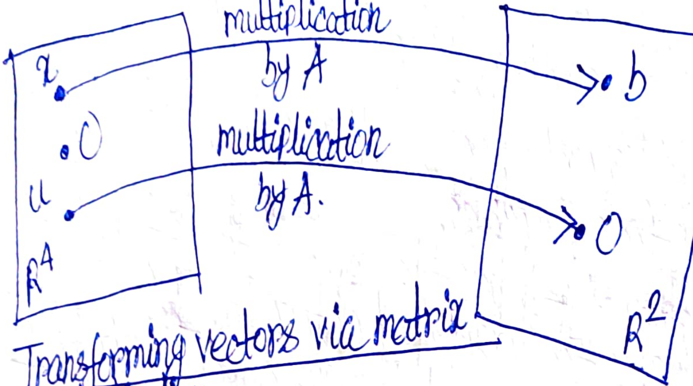

Transforming vectors via matrix multiplication

- From this new point of view, solving the equation $Ax=b$ amounts to finding all vectors x in R^4 that are transformed into the vector b in R^2 under the "action" of multiplication by A.

- A $\boxed{\text{transformation}}$ (or $\boxed{\text{function}}$ or $\boxed{\text{mapping}}$) T from R^n to R^m is a rule that assigns to each vector x in R^n a vector $T(x)$ in R^m. The set R^n is called the $\boxed{\text{domain}}$ of T, and R^m is called the $\boxed{\text{codomain}}$ of T. The notation $T: R^n \rightarrow R^m$ indicates that the domain of T is R^n and the codomain is R^m. For x in R^n, the vector $T(x)$ in R^m is called the $\boxed{\text{image of } x}$ (under the action of T). The set of all images $T(x)$ is called the $\boxed{\text{range}}$ of T. ↳ Imp point.

Matrix Transformations

- For each x in R^n, $T(x)$ is computed as Ax, where A is an $m \times n$ matrix.
$\boxed{\text{Matrix transformation}}$: $x \mapsto Ax$.
- Observe that the domain of T is R^n where A has n columns and the codomain of T is R^m when each column of A has m entries.
- The range of T is the set of all linear combinations of the columns of A, because each image $T(x)$ is of the form Ax.

$\boxed{\text{Example 1:}}$ Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$, and define a transformation $T: R^2 \rightarrow R^3$ by $T(x) = Ax$, so that

$$T(x) = Ax = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$